

Runway Incursion

Terms 1

Controlled Movement Area (CMA) - Any portion of the airfield requiring aircraft, vehicles, and pedestrians to obtain specific Air Traffic Control approval for access (normally via two-way radio contact with the Control Tower).

Note: Controlled Movement Areas include, but are not limited to areas used for takeoff, landing, and required taxiing of aircraft.

Controlled Movement Area Violation (CMAV) Event - An airfield infraction caused by aircraft, vehicles, or pedestrians entering the control movement area without specific Control Tower approval.

Note: This definition includes runway incursions and infractions caused by communication errors.



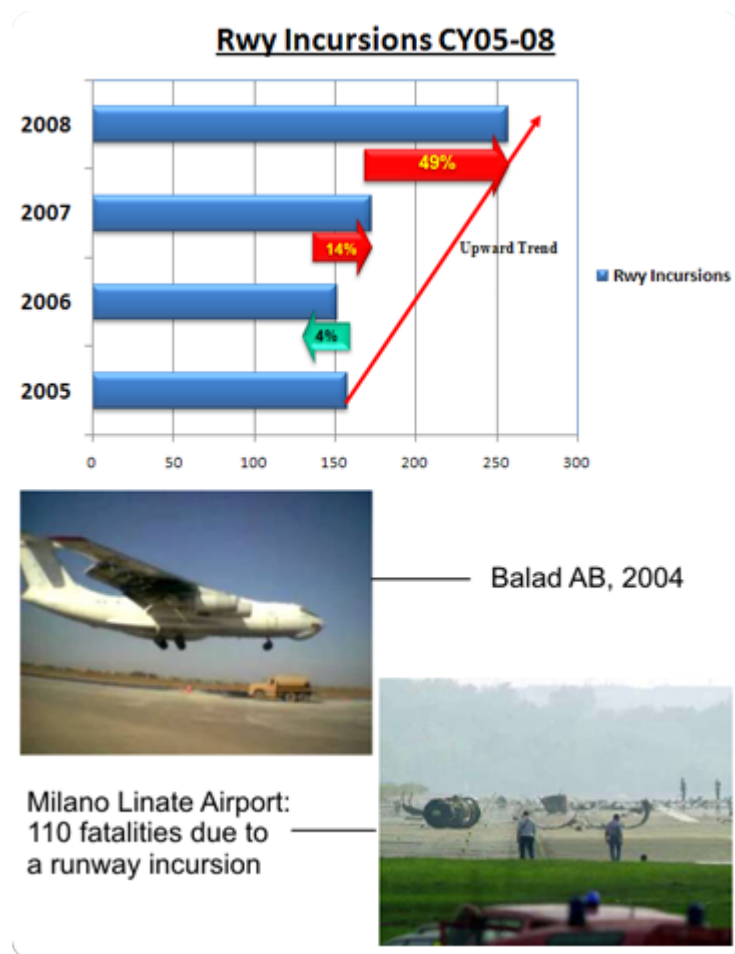
Terms 2

A runway incursion is one of the most serious airfield violations you as a vehicle driver can commit.

Runway Incursion - A Controlled Movement Area Violation (CMAV) caused by unauthorized entry or erroneous occupation of any surface used for takeoff and landing of aircraft, regardless of impact on aircraft safety. These incidents can be caused by aircraft, vehicles, or pedestrians. There's potential for a runway incursion at every runway.

The Leading causes of runway incursions (Human Factors) are:

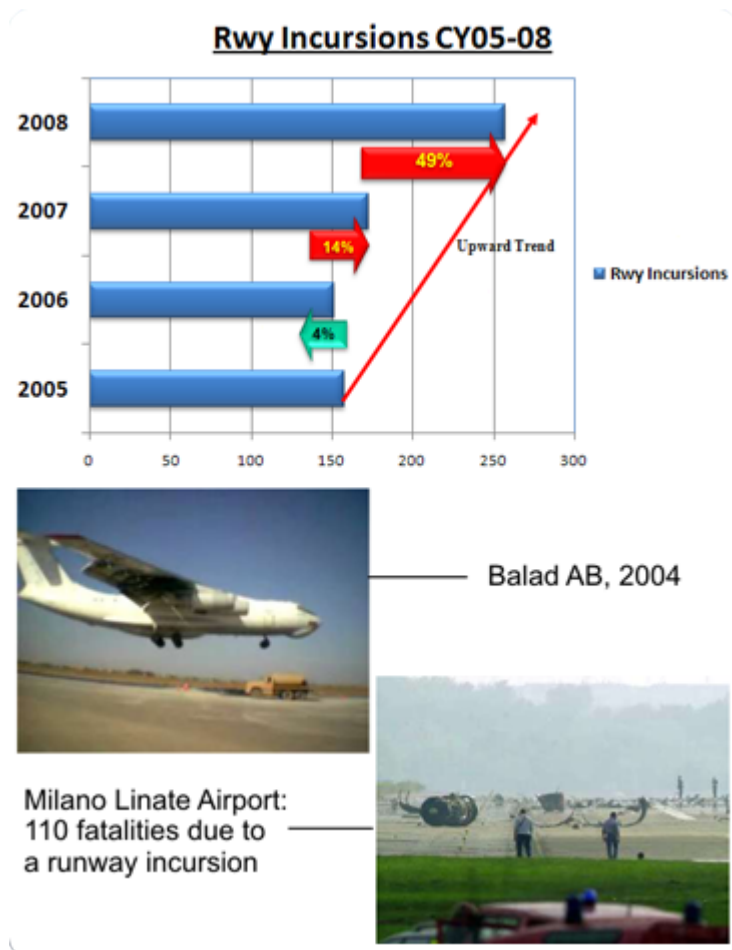
- Miscommunication
- Lack of situational awareness
- Insufficient training



Terms 3

Crossing the runway without ATC approval will result in:

- immediate apprehension.
- suspension of airfield driving privileges (average 30 days).
- letter to your commander.



Terms 4

Runway incursions are easily preventable and warrant serious repercussions if incurred. At best, your airfield driving privileges will be revoked and a hazard report will be initiated. At worst, your actions will result in the death of yourself and others and the destruction of valuable military assets.

Situational awareness (SA) is an essential component of the Runway Incursion Prevention Program. It is defined as having the knowledge and understanding of the airfield environment and/or airfield operations to maintain safe driving operations.

Maintain situational awareness by:

- Understanding and following Control Tower instructions
- Using airfield diagrams
- Knowing the purpose and requirements indicated by airfield markings, signs and lights



Terms 5

Understanding the terms associated with the Runway Incursion Prevention Program will help you develop your SA.

Remember, runway incursions are occurrences at an airfield involving the incorrect presence of an aircraft, vehicle or person on the protected area of a surface designated for the landing and takeoff of aircraft.

Keep your awareness high and prevent CMAV's like the example on the right.



Incidents 1

A large number of runway incursions occur during the response to an aircraft emergency. At all times during the aircraft emergency the runway is considered a Controlled Movement Area. Regardless of the aircraft status or your job (i.e. Rescue / Security / Aircraft Maintenance) you require Control Tower permission to enter the runway. The runway is still considered active.

Do not follow other emergency response vehicles on the runway, unless the Control Tower has directly given you permission to enter the runway or you have pre-arranged an escort with an authorized vehicle driver.



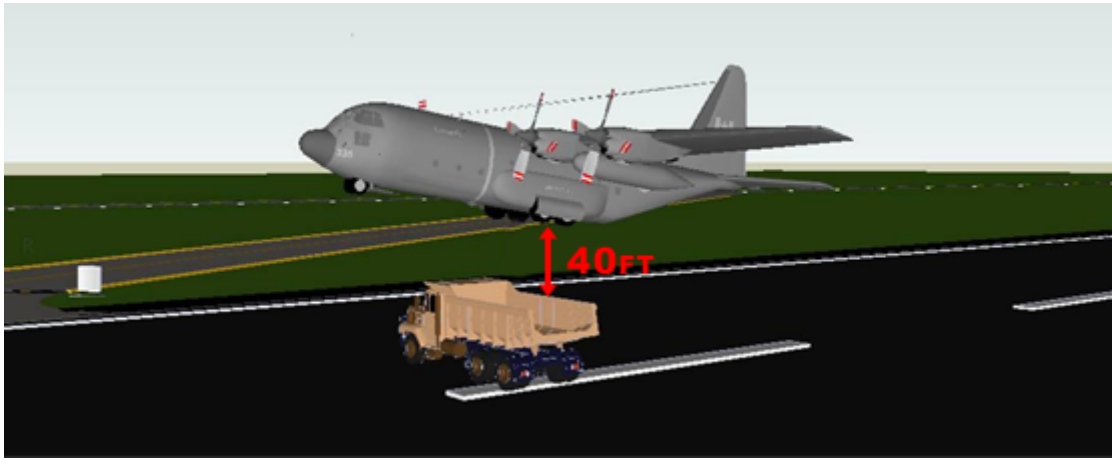
Incidents 2

Actual example of an extremely close call:

Sep 05 - The crew of a C-130 started take-off and kept an eye on the 3 trucks off the right side of the runway. At approximately 70 Knots into the take-off, they noticed one of the dump trucks moving slowly towards the runway, but still clear. The dump truck continued its forward movement, pulled onto the center of the runway and stopped perpendicular to the centerline.

The crew then realized that at their current speed of 90 Knots, they would not be able to stop the aircraft before impacting the dump truck. They elected to continue the take-off to clear the truck and used all the runway available to get maximum airspeed. They were forced to rotate 19 knots early to avoid impacting the truck.

The crew estimates that they cleared the truck by no more than 40 feet.



Incidents 3

Actual Runway Incursions:

Scenario 1

A Sweeper driver requested permission onto Runway 27 from Taxiway Bravo. The sweeper then proceeded across the hold line for Runway 26. There was a C-130 on a two mile final. The Sweeper crossed the wrong runway.

Scenario 2

A bobtail driver performing FOD reduction operations, (towing a FOD BOSS) was not in radio contact with the control tower and on the runway. The tower controller had to send a C-17 on final around. Bobtail driver did not have a radio and was improperly trained on airfield driving procedures.

Scenario 3

After an aircraft emergency the aircraft exited the runway with Fire Department vehicles following. Shortly thereafter, a blue Security Forces vehicle then entered the runway while another aircraft was on final. The aircraft was unable to go around and landed on the runway while the vehicle was driving down the centerline. Aircraft stopped within 4000 feet of vehicle. (Hazardous Air Traffic Report).

Summary

In this module you were introduced to the various components of Runway Incursion Prevention. You are now able to define various terms such as Situational Awareness (SA), Controlled Movement Area (CMA), and Controlled Movement Area Violations (CMAV).